

Modelling and analysis of Interleaved Cuk converter controlled by PID controller with phase shift PWM method

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Abstract—The increasing integration of the solar based system, EV's(Electrical vehicles) with the existing conventional generation has led to required dc-dc converters for interfacing. For improve the system performance the DC-DC converters have to meet many requirements, such as high energy density, less ripple in the system, low EMI(Electromagnetic interference), compact size. Hence, a DC-DC interleaved CUK converter (ICC) is most suitable choice for meeting the requirements. This paper is based on the novel idea of modeling and controlling an interleaved CUK converter. Modeling is carried by linearizing the nonlinear model of the converter via average and small signal modelling and tuning of controller parameters is based on heuristic search technique known as Genetic Algorithm. The results show a better response time (rise time, settling time) and error specification in case of GA tuned controller than without controller. Hence, it can be concluded that this element (converter) coupled controller can give the desired results as demanded by the energy device.

Index Terms—Interleaved converter, Cuk converter, Multi-phase system, PID controller, Genetic algorithm.

I. INTRODUCTION

Due to increased researches on energy studies, solar PV systems, fuel cells and hybrid electrical vehicles; the uses of the DC-DC converter is increasing. The voltages provided in these types of applications are low therefore, it is necessary to use excitation boost type topologies. Further, output ripples introduced by them should be less for good performance of the system. There are many DC-DC converter topologies present in the literature [1]- [4]. The CUK DC-DC converter has simultaneous property of step-up and step down the output voltage while making the input and output current of the converter continuous. In order to introduce less ripples in the system, coupled inductor can be used. In 1971 a multiphase topology was introduced by D. R. Garth et.al. [5] Thereby later, several improvements were introduced in making it efficient. such as implementation of phase shift concept on multiphase topology. The novel interleaved CUK Converter (ICC) is studied for the following advantages; step up and step down the output voltage, continuous input and output current, symmetry in structure, equal distribution of current in each

phase (this increases the load wearing and fault tolerance capability of the system), provide ripple cancellation which further reduces the ripple. This system has to be cascaded with a suitable controller for desired output. Any system, whether linear or nonlinear requires a controller for achieving its desired objective. The most commonly used controllers in either industrial or domestic purposes since 19th century are Proportional Integral Derivative (PID) controllers. The early applications of PID controllers in industry were used for designing governors [6] for controlling speed of machines. At present, PID controllers have been most significant controllers in all spheres of controller applications ranging from automation and control; motion control; power control etc. and accounting to 90% of the process industries controllers [7]. The extensive use of PID controller is due to its simple structure, ability to be applied upon by advanced control techniques and ease of re-tuning online. The first tuning rules of primitive controllers similar to PID (proportional derivative controller) were laid by Callender for a first order plus time delay system [8]. The prominent work done by Zeigler and Nicholes formed the basis of tuning of all the modern day PID controllers [9]. Cohen and Coon modified the previous method based on first order plus time delay (FOPDT) and gave a new designing technique of PID controller [10]. The first analytical method of synthesizing the controller for repeated poles based on Gain Margin (GM) and Phase Margin (PM) was carried out by Skoczowski and Tarasiejski [11]. Gradually along with conventional techniques of tuning, evolutionary designing algorithms started being developed. The earlier work of designing PID controller via evolutionary algorithms was proposed by Porter et.al. This work was based on Genetic Algorithm (GA) based synthesizing of parameters of the controller [12]. Wang et.al [13] and Ho et.al [14] proposed the optimal tuning of PID controller via GA based on gain and phase margin.

There are many natural algorithms of optimizations existing in nature today. Some of them are Genetic Algorithm (GA),

Particle Swarm Optimizations (PSO), Cuckoo Search Optimization (CSO), Ant Colony Optimization (ACO) [15] etc. John Holland worked on adaptive systems and later developed an approach of Genetic Algorithm. This is based on natural phenomenon and which follows the principle of Survival of the fittest. It starts with individual selection, followed by evaluation of fitness function, rejecting what is bad and cross combining or mutating what is good and can be made better [16].

This work discusses modelling and controlling of ICC based devices. For desired output operation, ICC needs to operate in closed loop with error compensator (controller). Since, the converter has four different modes of operation, accordingly its switching shows non-linear characteristics. Therefore, average modeling and small signal analysis is carried out for applying linear control approach on the nonlinear plant. Finally, the controller is designed and the parameter of PID controller is calculated via Genetic Algorithm (GA) optimization evaluating the fitness function as the weighted sum of error performances. Error performances used here are Integral Square Error (ISE), Integral Absolute Error and Integral Time of Absolute Error (ITAE). Discuss the servo, and regulatory performances of the system. The experimental result is taken in MATLAB Simulink.

This research paper is divided into following sections. Section I deals with the literature survey on PID controllers, the intelligent algorithm and the converters significance. Section II highlights the mathematical modeling and pre-requisites of the things discussed in the paper. Section III talks about the control approach that has been used in the paper, Section IV discusses about the simulation results and Section V concludes the full paper.

II. ICC(INTERLEAVED CUK CONVERTER)

The Fig. 1 shows the circuit diagram of the 2-phase ICC. It has two branch $L_{1a}, L_{1b}, C_{1a}, C_{1b}$ are the capacitors and inductor of the branch one and $L_{2a}, L_{2b}, C_{2a}, C_{2b}$ are the capacitors and inductor of the branch two. one output capacitor can also used instead of two capacitor (C_{1b}, C_{2b}).

A. Mathematical modelling of Interleaved cuk converter

The interleaved cuk converter has four mode of operation.

TABLE I
ICC MODES OF OPERATION

Modes	Sw1	Sw2	D1	D2
Modes-1	closed	open	open	closed
Modes-2	open	closed	closed	open
Modes-3	open	open	closed	closed
Modes-4	closed	closed	open	open

1) *Mode-I*: In Mode-I operation the switch Sw1 is closed and Sw2 is open and the diode D1 and D2 get open and closed respectively, due to that the inductor current i_{L1a} and i_{L2a} is exponentially increases and decreases respectively, similarly in i_{L1b} and i_{L2b} . The capacitor voltage v_{C1a} get

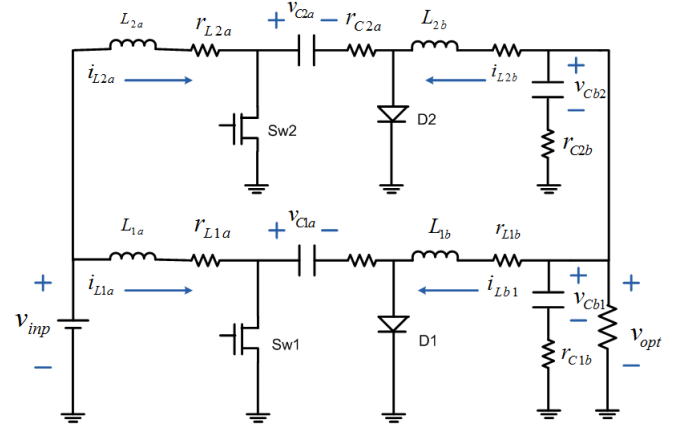


Fig. 1. Circuit diagram of 2-phase ICC

exponentially discharge and v_{C2a} get charge. The system state space equation in this mode are as follow:

$$\frac{di_{L1a}}{dt} = \frac{V_s}{L_{1a}} - \frac{(r_{L1a} - r_{sw})}{L_{1a}} i_{L1a} - \frac{r_{sw}}{L_{1a}} i_{L1b} \quad (1)$$

$$\frac{di_{L1b}}{dt} = -\frac{r_{sw}}{L_{1b}} i_{L1a} - \frac{R_f}{L_{1b}} i_{L1b} + \frac{v_{C1a}}{L_{1b}} - \frac{2Rv_{C1b}}{(2R + r_{C1b})L_{1b}} \quad (2)$$

$$\frac{dv_{C1a}}{dt} = -\frac{i_{L1b}}{C_{1a}} \quad (3)$$

$$\frac{dv_{C1b}}{dt} = \frac{2R}{(2R + r_{C1b})C_{1b}} i_{L1b} - \frac{V_{C1b}}{(2R + r_{C1b})C_{1b}} \quad (4)$$

$$\frac{di_{L2a}}{dt} = -\frac{R_h}{L_{2a}} i_{L2a} - \frac{r_d}{L_{2a}} i_{L2b} - \frac{v_{C2a}}{L_{2a}} + \frac{V_s}{L_{2a}} \quad (5)$$

$$\frac{di_{L2b}}{dt} = -\frac{r_d}{L_{2b}} i_{L2a} - \frac{R_g}{L_{2b}} i_{L2b} - \frac{2R}{(2R + r_{C2b})} v_{C2b} \quad (6)$$

$$\frac{dv_{C2a}}{dt} = \frac{i_{L2a}}{C_{2a}} \quad (7)$$

$$\frac{dv_{C2b}}{dt} = \frac{2R}{(2R + r_{C2b})C_{2b}} i_{L2b} - \frac{v_{C2b}}{(2R + r_{C2b})C_{2b}} \quad (8)$$

2) *Mode-2*: In Mode-2 operation the switch Sw1 is open and Sw2 is closed and the diode D1 and D2 get closed and open respectively, due to that the inductor current i_{L1a} and i_{L2a} is exponentially decreases and increases respectively, similarly in i_{L1b} and i_{L2b} . The capacitor voltage v_{C1a} get exponentially discharge and v_{C2a} get charge. The system state space equations in this mode are as follow:

$$\frac{di_{L1a}}{dt} = -\frac{R_h}{L_{1a}} i_{L1a} - \frac{r_d}{L_{1a}} i_{L1b} - \frac{v_{C1a}}{L_{1a}} + \frac{V_s}{L_{1a}} \quad (9)$$

$$\frac{di_{L1b}}{dt} = -\frac{r_d}{L_{1b}} i_{L1a} - \frac{R_g}{L_{1b}} i_{L1b} - \frac{2R}{(2R + r_{C1b})} v_{C1b} \quad (10)$$

$$\frac{dv_{C1a}}{dt} = \frac{i_{L1b}}{C_{1a}} \quad (11)$$

$$\frac{dv_{C1b}}{dt} = \frac{2R}{(2R + r_{C1b})C_{1b}} i_{L1b} - \frac{v_{C1b}}{(2R + r_{C1b})C_{1b}} \quad (12)$$

$$\frac{di_{L2a}}{dt} = \frac{V_s}{L_{2a}} - \frac{(r_{L2a} - r_{sw})}{L_{2a}} i_{L2a} - \frac{r_{sw}}{L_{2a}} i_{L2b} \quad (13)$$

$$\frac{di_{L2b}}{dt} = -\frac{r_{sw}}{L_{2b}} i_{L2a} - \frac{R_f}{L_{2b}} i_{L2b} + \frac{v_{C2a}}{L_{2b}} - \frac{2Rv_{C2b}}{(2R + r_{C2b})L_{2b}} \quad (14)$$

$$\frac{dv_{C2a}}{dt} = -\frac{i_{L2b}}{C_{2a}} \quad (15)$$

$$\frac{dv_{C2b}}{dt} = \frac{2R}{(2R + r_{C2b})C_{2b}} - \frac{V_{C2b}}{(2R + r_{C2b})C_{2b}} \quad (16)$$

3) *Mode-3*: In Mode-3 operation the switch Sw1 and Sw2 get open and the diode D1 and D2 get closed respectively, due to that the inductor current i_{L1a} and i_{L2a} is exponentially decreases, similarly in i_{L1b} and i_{L2b} . The capacitor voltage v_{C1a} and v_{C2a} get exponentially charge. The system state space equations in this mode are as follow:

$$\frac{di_{L1a}}{dt} = -\frac{R_h}{L_{1a}} i_{L1a} - \frac{r_d}{L_{1a}} i_{L1b} - \frac{v_{C1a}}{L_{1a}} + \frac{V_s}{L_{1a}} \quad (17)$$

$$\frac{di_{L1b}}{dt} = -\frac{r_d}{L_{1b}} i_{L1b} - \frac{R_g}{L_{1b}} i_{L1b} - \frac{2R}{(2R + r_{C1b})} v_{C1b} \quad (18)$$

$$\frac{dv_{C1a}}{dt} = \frac{i_{L1b}}{C_{1a}} \quad (19)$$

$$\frac{dv_{C1b}}{dt} = \frac{2R}{(2R + r_{C1b})C_{1b}} i_{L1b} - \frac{v_{C1b}}{(2R + r_{C1b})C_{1b}} \quad (20)$$

$$\frac{di_{L2a}}{dt} = -\frac{R_h}{L_{2a}} i_{L2a} - \frac{r_d}{L_{2a}} i_{L2b} - \frac{v_{C2a}}{L_{2a}} + \frac{V_s}{L_{2a}} \quad (21)$$

$$\frac{di_{L2b}}{dt} = -\frac{r_d}{L_{2b}} i_{L2a} - \frac{R_g}{L_{2b}} i_{L2b} - \frac{2R}{(2R + r_{C2b})} v_{C2b} \quad (22)$$

$$\frac{dv_{C2a}}{dt} = \frac{i_{L2a}}{C_{2a}} \quad (23)$$

$$\frac{dv_{C2b}}{dt} = \frac{2R}{(2R + r_{C2b})C_{2b}} i_{L2b} - \frac{v_{C2b}}{(2R + r_{C2b})C_{2b}} \quad (24)$$

4) *Mode-4*: In Mode-4 operation the switch Sw1 and Sw2 get closed and the diode D1 and D2 get open, due to that the inductor current i_{L1a} and i_{L2a} is exponentially increases, similarly in i_{L1b} and i_{L2b} . The capacitor voltage v_{C1a} and v_{C2a} get discharge exponentially. The system state space equations in this mode are as follow:

$$\frac{di_{L1a}}{dt} = \frac{V_s}{L_{1a}} - \frac{(r_{L1a} - r_{sw})}{L_{1a}} i_{L1a} - \frac{r_{sw}}{L_{1a}} i_{L1b} \quad (25)$$

$$\frac{di_{L1b}}{dt} = -\frac{r_{sw}}{L_{1b}} i_{L1a} - \frac{R_f}{L_{1b}} i_{L1b} + \frac{v_{C1a}}{L_{1b}} - \frac{2Rv_{C1b}}{(2R + r_{C1b})L_{1b}} \quad (26)$$

$$\frac{dv_{C1a}}{dt} = -\frac{i_{L1b}}{C_{1a}} \quad (27)$$

$$\frac{dv_{C1b}}{dt} = \frac{2R}{(2R + r_{C1b})C_{1b}} i_{L1b} - \frac{V_{C1b}}{(2R + r_{C1b})C_{1b}} \quad (28)$$

$$\frac{di_{L2a}}{dt} = \frac{V_s}{L_{2a}} - \frac{(r_{L2a} - r_{sw})}{L_{2a}} i_{L2a} - \frac{r_{sw}}{L_{2a}} i_{L2b} \quad (29)$$

$$\frac{di_{L2b}}{dt} = -\frac{r_{sw}}{L_{2b}} i_{L2a} - \frac{R_f}{L_{2b}} i_{L2b} + \frac{v_{C2a}}{L_{2b}} - \frac{2Rv_{C2b}}{(2R + r_{C2b})L_{2b}} \quad (30)$$

$$\frac{dv_{C2a}}{dt} = -\frac{i_{L2b}}{C_{2a}} \quad (31)$$

$$\frac{dv_{C2b}}{dt} = \frac{2R}{(2R + r_{C2b})C_{2b}} - \frac{V_{C2b}}{(2R + r_{C2b})C_{2b}} \quad (32)$$

The equation (1) - (32) shows the state space equation of the interleaved cuk converter in a period. For apply the control technique required to derive the average model and linearized it. For averaging and linearizing, average modeling and small signal modeling is use.

B. Average and small signal modeling

The above equation give the four sets of state space matrix, let the transition matrix is A_1, A_2, A_3, A_4 , input matrix is B_1, B_2, B_3, B_4 and the output matrix is C_1, C_2, C_3, C_4 . The state space matrix of average model is:

For $d > 0.5$

$$A = (1 - d)(A_1 + A_2) + (2d - 1)A_4 \quad (33)$$

$$B = (1 - d)(B_1 + B_2) + (2d - 1)B_4 \quad (34)$$

$$C = (1 - d)(C_1 + C_2) + (2d - 1)C_4 \quad (35)$$

For $d < 0.5$

$$A = d(A_1 + A_2) + (1 - 2d)A_3 \quad (36)$$

$$B = d(B_1 + B_2) + (1 - 2d)B_3 \quad (37)$$

$$C = d(C_1 + C_2) + (1 - 2d)C_3 \quad (38)$$

Now, the average state space model of the system,

$$\dot{x} = Ax + Bu \quad (39)$$

$$y = Cx \quad (40)$$

Where, d is the duty ration, x is the state variable and u is the input of the converter. For small signal modeling of the system replace the following value in the equation (39) and (40).

$$x = X + \hat{x} \quad u = U + \hat{u}$$

$$d = D + \hat{d} \quad y = Y + \hat{y}$$

Here, the capital value shows the steady state value of the system, the letter with cap shows the variation in the value and small letter value shows the intantenous value.

For $d < 0.5$

$$\begin{aligned} \dot{X} + \hat{x} = & ((D + \hat{d})(A_1 + A_2) + (1 - 2(D + \hat{d}))A_3)(X + \hat{x}) \\ & + ((D + \hat{d})(B_1 + B_2) + (1 - 2(D + \hat{d}))B_3)(U + \hat{u}) \end{aligned} \quad (41)$$

$$\hat{Y} + \hat{y} = ((D + \hat{d})(C_1 + C_2) + (1 - 2(D + \hat{d}))C_3)(X + \hat{x}) \quad (42)$$

Here steady state term and the term with two cap value is neglected. On simplifying and re-arranging the above equation yeild as,

$$\dot{\hat{x}} = A''\hat{x} + B''\hat{d} \quad (43)$$

$$\hat{y} = C''\hat{x} + D''\hat{d} \quad (44)$$

Where,

$$A'' = (A_3 + (A_1 + A_2 - 2A_3))$$

$$B'' = ((A_1 + A_2 - 2A_3)X + (B_1 + B_2 - 2B_3)U)$$

$$C'' = C_3 + (C_1 + C_2 - 2C_3)D$$

$$D'' = (C_1 + C_2 - 2C_3)X$$

On laplace transform of the equation (43),(44) and arranging them yeild as,

$$\hat{y}(s) = (C''(sI - A)^{-1}B'' + D'')\hat{d}(s) \quad (45)$$

or

$$\frac{\hat{y}(s)}{\hat{d}(s)} = (C''(sI - A)^{-1}B'' + D'') \quad (46)$$

This equation shows the transfer function of the converter for $d > 0.5$.

The equation shows the transfer functions of the converter when design parameter shows in table II is put in the equation (46).

$$G(s) = \frac{-8550s^3 + 1.354 \times 10^8 s^2 - 1.08 \times 10^{11} s + 1.871 \times 10^{14}}{s^4 + 507.9s^3 + 8.38 \times 10^6 s^2 + 1.233 \times 10^9 s + 3.288 \times 10^{12}} \quad (47)$$

similarly yeild for $d < 0.5$.

$$G(s) = \frac{-8550s^3 + 1.354 \times 10^8 s^2 - 1.063 \times 10^{11} s + 4.22 \times 10^{14}}{s^4 + 507.9s^3 + 8.38 \times 10^6 s^2 + 1.233 \times 10^9 s + 3.288 \times 10^{12}} \quad (48)$$

TABLE II
DESIGN SPECIFICATIONS OF ICC

Converter Parameter	Value	unit
Source voltage	12	V
frequency	20	kHz
Duty ration	0.6	-
Load resistance	9	ω
Inductor L_{1a}, L_{2a}	3245	μH
Inductor L_{1b}, L_{2b}	2160	μH
Capacitor C_{1a}, C_{2a}	90	μF
Capacitor C_{1b}, C_{2b}		μF
Output current ripple	5%	
Output voltage ripple	5%	

III. CONTROLLER DESIGN

A. PID controller

A simple structure of PID controller consists of three parameters namely Proportional gain (K_p), Integral gain (K_i) and Derivative gain (K_d). The transfer function of PID controller is given by $C(s)$.

$$C_{PID}(s) = K_P + \frac{K_I}{s} + K_D s \quad (49)$$

Thus, tuning the above said parameters according to the systems desired performance is very essential for its proper functioning.

The interleaved converter is modeled as given in equation (48) and PID parameters are calculated by GA.

PID controller transfer function obtained

$$C_{PID}(s) = 0.0029 + \frac{0.8981}{s} + 6.1544 \times 10^{-6} s \quad (50)$$

B. Genetic algorithm

Genetic algorithm works around the following cycle:

- **Initialization of population:** A sample of population or chromosomes is selected randomly and digitised into sequence of binary digits known as strings.
- **Selection and Rejection:** The converted strings are evaluated through fitness function and are either selected or rejected on the basis of their evaluation.
- **Reproduction:** The selected chromosomes are combined to produced new offsprings via crossover of string digits o produce a better generation than their parents. Further, mutation of offsprings are also carried out to better the succeeding generations.

This process continues until the last generation converges to the optimized value (maximum or minimum).

1) *Control Problem:* After modelling the interleaved converter, the tuning of PID controlled system parameters have been carried out by GA optimization minimizing the cost function as weighted sum of error performance specifications

$$J = \omega_1 \int_0^\infty t|e|dt + \omega_2 \int_0^\infty e^2 dt + \omega_3 \int_0^\infty |e|dt \quad (51)$$

where e , t , ω_i ; $i = 0, 1, 2$, are the feedback error, total simulation time and the weights respectively. The bounds in the GA problem is set by hit and trial method of PID tuning via PID tuner application in MATLAB [17].

IV. SIMULATION RESULTS AND CONTROL ASPECTS

In this manuscript the control response and experimental results is find for a duty cycle < 0.5 .

A. Control Aspects

1) *Servo and Regulatory responses:* The servo response and regulatory responses (perturbation given to the system at 2 seconds of 0.5 amplitude) of non-controlled closed loop feedback system and PID controlled feedback system is compared in Fig. (2) and Fig. (3) and summed in Table III respectively.

From Fig. (2), Fig. (3) and Table III, it can be inferred that PID controlled system shows better optimal desired responses in both responses of servo and regulatory with respect to t_r , t_s , and IAE.

2) *Relative stability :* The frequency plots of magnitude and phase for system without and with controller respectively are plotted in Fig. (4).

From Fig. (4) it can be concluded that plant with PID controller is relatively more stable than closed loop system without the PID controller. From Fig. (4) it can be concluded that plant with PID controller is relatively more stable than closed loop system without the PID controller.

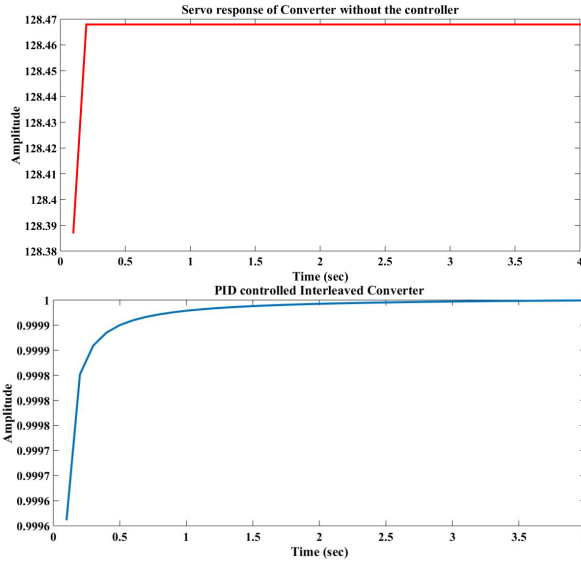


Fig. 2. Servo responses of the plant without controller (red) and with controller (blue)

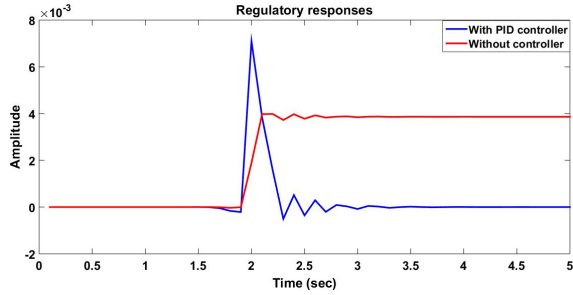


Fig. 3. Regulatory responses of the plant without controller (red) and with controller (blue)

TABLE III
COMPARISON OF SERVO AND REGULATORY RESPONSES FOR THE SYSTEM WITHOUT AND WITH CONTROLLER

Servo Response			
Controller	Rise time	settling time	IAE
No controller	0.1209	1.9865	0.1699
PID	0.2526	1.2955	0.01468
Regulatory Responses			
Controller	Rise time	settling time	IAE
No Controller	0.040	2.808	0.0093
PID	0.189	2.721	0.0015

B. Simulation results

The ICC simulation is performed in MATLAB simulink environment. The simulation is perform for the duty cycle $D < 0.5$. Here the 50% phase shift is taken because of relation $\frac{2\pi}{N}$ (N is number of phase in this case $N=2$) Fig.(5)-(10) shows the results at constant 12 V input. The Fig. (5) shows the inductor L_{1a} , L_{2a} current, The Fig. (6) shows the inductor L_{1b} , L_{2b} current, Fig. (7) shows the capacitor C_{1b} , C_{2b} voltage, Fig. (8), Fig. (9) shows the input and output current respectively and the Fig. (10) shows the output voltage of ICC.

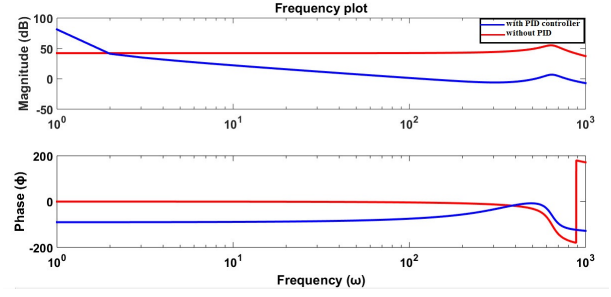


Fig. 4. Frequency response analysis of system without and with controller

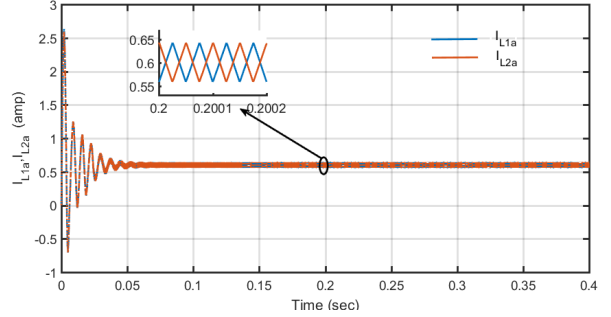


Fig. 5. Inductor L_{1a} , L_{2a} current at constant source voltage

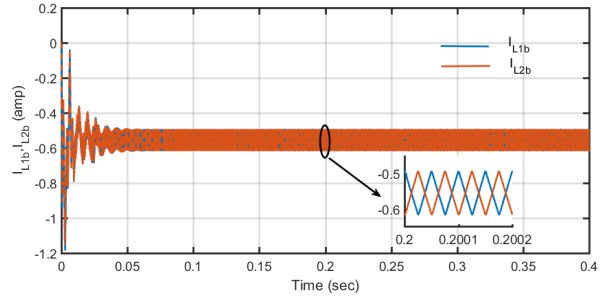


Fig. 6. Inductor L_{1b} , L_{2b} current at constant source voltage

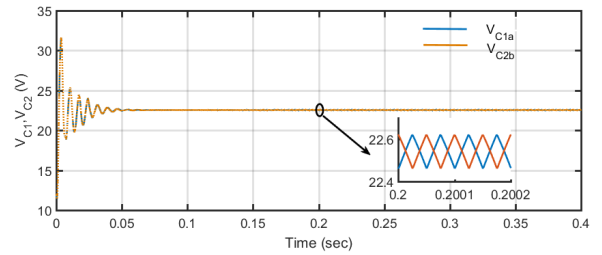


Fig. 7. Capacitor C_{1b} , C_{2b} voltage at constant source voltage

V. CONCLUSION

This paper discusses about the energy based converter interleaved CUK converter and tries to model and design a PID controller through a novel approach known as Genetic

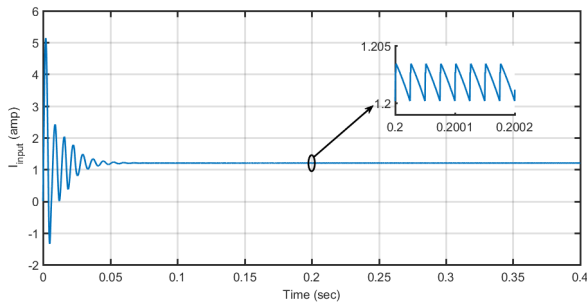


Fig. 8. input current of ICC at constant source voltage

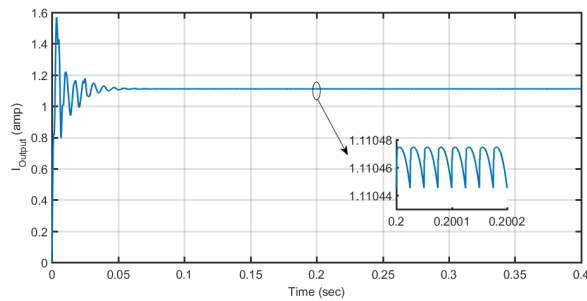


Fig. 9. Output current of ICC at a constant source voltage

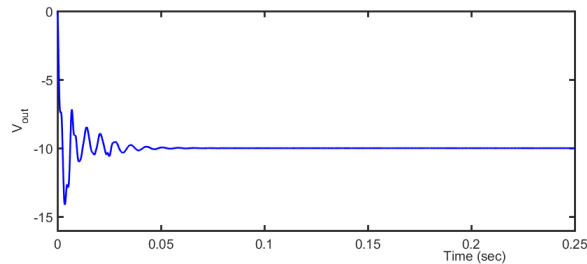


Fig. 10. Output voltage of ICC at a constant source voltage

Algorithm. The results shows a much better improved response with lesser rise time, settling time and desirable ripples when PID controller is installed in the system as compared to when the system is non-controlled. The ripple value observe from the simulation is less than 1% which is with in the permissible limit.

REFERENCES

- [1] L. H. S. C. Barreto, P. P. Praca, D. S. O. Jr and R. P. T. Bascope, "Single-stage topologies integrating battery charging, high voltage step-up and photovoltaic energy extraction capabilities," in *Electronics Letters*, vol. 47, no. 1, pp. 49-50, 6 January 2011.
- [2] Y. Hsieh, J. Chen, T. Liang and L. Yang, "Novel High Step-Up DCDC Converter for Distributed Generation System," in *IEEE Transactions on Industrial Electronics*, vol. 60, no. 4, pp. 1473-1482, April 2013.
- [3] S. Chen, T. Liang, L. Yang and J. Chen, "A Cascaded High Step-Up DCDC Converter With Single Switch for Microsource Applications," in *IEEE Transactions on Power Electronics*, vol. 26, no. 4, pp. 1146-1153, April 2011.
- [4] M. H. Todorovic, L. Palma and P. Enjeti, "Design of a wide input range DC-DC converter with a robust power control scheme suitable for fuel cell power conversion," *Nineteenth Annual IEEE Applied Power Electronics Conference and Exposition*, 2004. APEC '04., Anaheim, CA, USA, 2004, pp. 374-379 Vol.1.
- [5] D. R. Garth, W. J. Muldoon, G. C. Benson and N. Costague, "Multi-phase, 2-kilowatt, high-voltage, regulated power supply," *1971 IEEE Power Electronics Specialists Conference*, Pasadena, CA, 1971, pp. 110-116.
- [6] S. Bennett, "Nicholas Minorsky and the automatic steering of ships," in *IEEE Control Systems Magazine*, vol. 4, no. 4, pp. 10-15, November 1984.
- [7] Araki, M. "PID control. *Control Systems, Robotics and Automation: System Analysis and Control: Classical Approaches II*," pp. 58-79, Unbehauen, H., 2009.
- [8] A. Callender, Douglas Rayner Hartree, and A. Porter, "Time-lag in a control system," *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, vol. 235, no. 756, pp. 415-444, July 1936.
- [9] J. G. Ziegler and N. B. Nichols, "Optimum settings for automatic controllers," *Transactions of ASME*, vol. 64, pp. 759-768, 1942.
- [10] C. Coon, "Theoretical Consideration of Retarded Control, *Transactions of the ASME*, Vol. 75, pp. 827-834, 1952.
- [11] Skoczowski, S., and L. Tarasiejski, "Tuning of PID controllers based on gain and phase margin specifications using Strejcs process model with time delay." In *Proceedings of the Third International Symposium on Methods and Models in Automation and Robotics (MMAR96)*, Miedzyzdroje, Poland, pp. 765-770, 1996.
- [12] B. Porter, A. H. Jones, "Genetic Tuning of Digital PID Controllers" *IEEE Electronics Letters*, Vol. 28, pp. 843-844, Apr 1992.
- [13] P. Wang and D. P. Kwok, "Optimal design of PID process controllers based on genetic algorithms," *Control Engineering Practice*, vol. 2, no. 4, pp. 641-648, 1994.
- [14] Aniruddha Datta, Ming-Tzu Ho, and Shankar P. Bhattacharyya, "Structure and synthesis of PID controllers," Springer Science & Business Media, 2013.
- [15] Dorigo, Marco, Mauro Birattari, Christian Blum, Maurice Clerc, Thomas Sttze, and Alan Winfield, "Ant Colony Optimization and Swarm Intelligence," *6th International Conference, ANTS 2008, Brussels, Belgium*, September 22-24, 2008.
- [16] Davis, Lawrence. "Handbook of genetic algorithms," 1991.
- [17] 2018.MathWorks.https://www.mathworks.com/tagteam/86589_pid-controller-tuning-in-simulink.pdf.